FastDedup

A fast and memory efficient tool for reads deduplication

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Abstract – TODO

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1 – Context

With the rise of New Generation Sequencing (NGS) technologies, the volume of sequencing data has increased exponentially. With this surge in data, the development of new tools for processing and analyzing sequencing reads was needed. One of the first and critical step in the analysis pipeline is the deduplication of reads, which involves identifying and removing duplicate sequences that can arise from PCR amplification or sequencing errors.

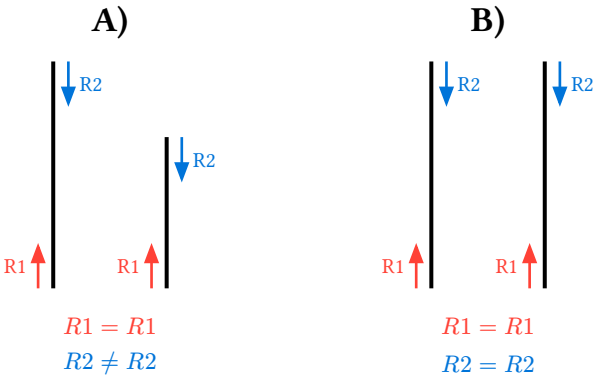


Figure 1: Identifying PCR Duplicates in Paired-End Sequencing.

Solid Black Line: Represents a single sequenced DNA fragment (height reflects length).

Red Arrow (R1): Forward read (Read 1).

Blue Arrow (R2): Reverse read (Read 2).

A) Not a duplication: Shows two distinct fragments where the start position of Read 1 is shared, but different fragment lengths lead to unique mapping positions for Read 2.

B) PCR Duplication: Shows two exact fragments where the start position of Read 1 and Read 2 is shared.

We define a fragment F, the 5′->3′ sequence as $R1$ with a length of r_1 and the 3′->5′ sequence as $R2$ with a length of r_2. We can say that $R1$ is the prefix of F, $R2$ reversed is the suffix of F and $r_1 = r_2$.

Therefore, a PCR duplicate can be defined as if F_1 and F_2 have the same starting and ending position in the genome, with the same length. In that case, $R1_1$ and $R1_2$ are the same, and $R2_1$ and $R2_2$ are the same.

This step is crucial because removing these artifacts preventing false-positive during variant calling since PCR polymerases can introduce errors during the cycles of amplification. If an error occurs early, it is propagated into multiple daughter molecules. When sequenced, these identical, error-carrying molecules will appear as multiple independent reads supporting a mutation that does not actually exist in the biological sample [1].

Deduplication also allows correcting the allele frequency bias since not all DNA molecules amplify with equal efficiency due to factors like GC content or fragment length. This uneven amplification can artificially inflate the coverage of certain alleles. And it is also improving *de novo* assembly, especially in whole-genome sequencing. Paired-end reads are mapped to estimate the order and intervening distance between contigs, duplicates can create false-positive connections between contigs or introduce conflicting connections. [2].

Before presenting our tool, we will first review the existing tools for reads deduplication and their limitations.

2 – Existing tools and limitations

2.1 : Fastp

`fastp` is a widely adopted, all-in-one FASTQ preprocessor written in C++ that performs quality control, adapter trimming, and deduplication within a single pass [3,4]. Unlike traditional alignment-based tools that require computationally expensive mapping and coordinate sorting, `fastp` operates directly on raw sequences.

To achieve high performance, `fastp` implements a probabilistic deduplication mechanism based on Bloom filters and multiple hash arrays. For each read (or read pair), the sequence is mapped to multiple integer hash values. If all corresponding positions in the hash arrays are already marked as positive, the read is flagged as a duplicate. This streaming, single-pass architecture avoids reading and writing intermediate files to disk, drastically reducing I/O bottlenecks and resulting in ultra-fast execution speeds.

A significant advantage of `fastp` is its highly configurable memory management. Users can adjust the `--dup_calc_accuracy` parameter (ranging from level 1 to 6) to scale the hash buffer size from 1 GB up to 24 GB, allowing the tool to run on both standard personal computers and High-performance Computing (HPC) clusters.

However, this probabilistic approach involves a minor trade-off in absolute precision. `fastp` has a theoretical false-positive (hash collision) rate of approximately 0.01%, which, while negligible for most routine downstream analyses, represents a conceptual difference from exact deterministic methods. Furthermore, `fastp`'s deduplication algorithm relies heavily on the exact matching of sequence lengths and coordination regions. Consequently, applying certain quality trimming operations (such as sliding-window trimming via `--cut_front` or `--cut_tail`) prior to deduplication can interfere with the duplicate identification process, as trimmed reads may no longer perfectly match their duplicate counterparts.

2.2 : FastUniq

`FastUniq` [5] is a deterministic deduplication tool explicitly designed for paired-end (PE) short reads. By directly comparing the nucleotide sequences of read pairs, it provides an exact identification of PCR duplicates.

Algorithmically, `FastUniq` operates through a strict three-step pipeline:
1. loads the entirety of the paired reads into memory
2. sorts them based on their sequences using a merge sort algorithm
3. identifies duplicates by sequentially comparing adjacent read pairs in the sorted list.

A notable feature of its comparison logic is its ability to handle reads of varying lengths: a shorter sequence is considered identical to a longer one if it perfectly matches its 5′ prefix. If duplicates are found, the pair containing the longest reads is retained.

While `FastUniq` guarantees absolute precision for exact duplicate removal (zero mismatches allowed), its architectural design introduces critical resource bottlenecks. Because it relies on importing the entire dataset into RAM simultaneously using a hierarchical storage structure, its memory footprint scales linearly with the input file size. Benchmarking studies have highlighted this as a severe limitation for modern high-throughput data; for instance, processing a 538 GB Hi-C dataset required up to 1 TB of RAM [6].

Consequently, while highly accurate, `FastUniq` is often impractical for massive datasets without access to HPC clusters. Furthermore, its scope is strictly limited to paired-end data, it does not tolerate sequence mismatches, and its deduplication output is known to be sensitive to the order of the input files.

2.3 : Clumpify

`Clumpify`, a component of the Java-based BBTools suite [7], introduces a distinct algorithmic paradigm to duplicate removal based on k-mer grouping. Rather than relying on exact sequence hashing or global

coordinate sorting, `Clumpify` identifies reads that share common k-mers and groups them into localized clusters or "clumps".

This spatial reorganization provides a unique secondary benefit. By physically placing highly similar sequences adjacent to one another in the output file, `Clumpify` dramatically improves standard `gzip` compression efficiency, as the compression algorithm can utilize much closer reference pointers.

A major functional advantage of `Clumpify` is its high sensitivity to optical duplicates. By parsing the flowcell tile and pixel coordinates embedded within Illumina FASTQ headers, `Clumpify` (via parameters such as `dupedist`) can explicitly distinguish between amplification-derived PCR duplicates and sensor-derived optical duplicates. For paired-end data, `Clumpify` predominantly uses the first read (R1) to determine the clump, allowing the paired read (R2) to passively follow its counterpart, which maintains file synchronization while minimizing computational overhead.

To manage memory, `Clumpify` implements a multi-phase strategy (KmerSplit and KmerSort). If the dataset exceeds available RAM, the tool can divide the data into temporary groups on the disk, sorting them individually before merging.

However, benchmarking shows that its reliance on the Java Virtual Machine (JVM) can lead to variable stability and high memory overhead on massive datasets. Sometimes it requires multi-threading just to maintain stability on large inputs like 500+ GB Hi-C datasets [6].

2.4 : SeqKit

`SeqKit` [8,9] is a comprehensive, cross-platform, and ultra-fast toolkit for FASTA and FASTQ file manipulation. Written in the Go, it benefits from efficient concurrency management and the ability to produce static binaries with no external dependencies, making it highly accessible and stable across different operating systems.

For PCR deduplication, SeqKit provides the `rmdup` subcommand. This tool relies on an exact sequence matching approach using a hashing strategy. To process massive datasets without exhausting system memory, `seqkit rmdup` offers an MD5 mode (invoked via the `-m` or `--md5` flag). In this mode, the algorithm computes and stores only the MD5 digest of each sequence in the RAM rather than the full nucleotide sequence. This extreme memory efficiency allows SeqKit to maintain a very low resource footprint, even outperforming several newer tools when handling vast amounts of single-end (SE) data.

Despite its speed and memory efficiency, `SeqKit rmdup` has a notable architectural limitation: it does not natively support the synchronized deduplication of paired-end reads. The `rmdup` command evaluates and processes individual files independently. It will check duplicates inside each file separately, so if $R1_1 = R1_2$ but $R2_1 \neq R2_2$, $R1_2$ in will be flagged as a duplicate and removed, while $R2_2$ will be retained, leading to loss of data and desynchronization of the paired files. While workarounds exist using other SeqKit subcommands (such as `seqkit pair` to resynchronize files), `SeqKit rmdup` is generally considered the optimal choice strictly for single-end libraries.

2.5 : CD-HIT-DUP

`CD-HIT-DUP`, a specialized utility within the broader CD-HIT package [10,11], is designed to identify and remove both identical and nearly-identical duplicates. Unlike tools restricted to exact string matching, `CD-HIT-DUP` is particulary robust against sequencing errors. It incorporates a greedy incremental clustering algorithm that tolerates a user-defined threshold of mismatches, including both substitutions and insertions/deletions.

`CD-HIT-DUP` relies on a prefix-suffix comparison approach. The tool first bins reads that share an identical prefix of k nucleotides. To accelerate this binning and sorting process, it uses a dual numeric encoding strategy (initially base-5, then base-10), allowing prefixes of up to 27 nucleotides to be compactly represented as 64-bit integers. Within each bin, sequences are sorted by decreasing length, with the longest sequence acting as the cluster's seed. The algorithm then evaluates the suffixes of the remaining sequences against this

seed, utilizing a k-mer word-based index table to statistically predict sequence identity and bypass computationally expensive full pairwise alignments when possible. For paired-end libraries, `CD-HIT-DUP` conceptually joins the read pairs into a single sequence and checks prefixes at both ends simultaneously to ensure paired-end integrity.

While its error tolerance provides a nuanced, high-sensitivity approach to duplicate removal, `CD-HIT-DUP` introduces a severe computational bottleneck regarding memory consumption. Because the greedy clustering algorithm must retain representative sequences and cluster bins in RAM, its memory footprint scales aggressively with dataset size and the allowed mismatch threshold. Benchmarking has shown that processing a relatively modest 50 million single-end read library can consume between 33 GB and 53 GB of RAM. For massive modern datasets, such as a 538 GB Hi-C dataset, `CD-HIT-DUP` required approximately 1 TB of RAM, often crashing on resource-constrained hardware [6].

Furthermore, practical deployment of `CD-HIT-DUP` in modern automated workflows requires caution. The tool has known stability issues in containerized environments (e.g., Docker and Singularity), occasionally throwing segmentation faults or I/O assertion errors when inputs are provided via process substitution from compressed files.

2.6 : PRINSEQ++

`PRINSEQ++` [12] is a highly optimized C++ implementation of the Perl-based `prinseq-lite` program. It is designed to perform a comprehensive suite of quality control, filtering, trimming, and reformatting tasks for genomic and metagenomic sequence data. By utilizing POSIX threads (pthreads) for parallelization, it achieves processing speeds more than 10 to 16 times faster than its predecessor while maintaining a relatively low memory footprint.

For sequence deduplication, `PRINSEQ++` departs from exact string comparison or sorting heuristics by implementing a probabilistic data structure known as a Bloom filter. During execution, each sequence is transformed using multiple fast, non-cryptographic hash functions, and the corresponding indices in a shared bit-array are flagged. If a newly evaluated sequence maps entirely to bits that are already set to positive, it is classified as a duplicate. This probabilistic model is highly advantageous for multi-threaded environments because it enables asynchronous read and write access to the filter with minimal thread blocking.

Additionally, `PRINSEQ++` natively supports the direct reading and writing of `gzip`-compressed FASTQ and FASTA files without the need for intermediate disk I/O, which significantly reduces hard drive storage requirements. It also natively supports paired-end reads by processing pairs simultaneously and outputting synchronized R1 and R2 files.

However, the deduplication function (`-derep`) is strictly limited to exact duplicates and does not tolerate sequencing errors. Furthermore, the probabilistic nature of the Bloom filter introduces a negligible but theoretically non-zero rate of false positives. Finally, when `PRINSEQ++` is executed with multiple threads, the output sequences are not guaranteed to remain in the same order as the input files, which may require consideration if downstream tools expect strict global index synchronization.

2.7 : SAM/BAM ecosystem tools

In addition to tools that operate directly on raw FASTQ sequences, the bioinformatics ecosystem relies heavily on utilities designed for SAM and BAM formats. PCR deduplication tools like such as `samtools` (via `markdup`) [13] and Picard's `MarkDuplicates` [14].

While these tools are typically used downstream after mapping reads to a reference genome, it is technically possible to convert raw, unaligned FASTQ files directly into unmapped SAM/BAM files to process them through these pipelines. However, even without the computationally heavy step of genomic alignment, the sheer format conversion from FASTQ to SAM/BAM introduces significant and unnecessary I/O and processing overhead. Because our primary objective is to evaluate and develop a highly optimized, memory-efficient tool strictly for direct FASTQ-to-FASTQ deduplication, these format-dependent tools fall outside the scope of our comparative analysis.

3 – FastDedup: High-Performance FASTX Deduplication

The goal behind the development of `FastDedup` (or `FDedup` ; Figure 2 Figure 3) was to create a FASTX PCR deduplication tool that prioritizes maximum speed and memory efficiency. To achieve this, the tool relies on `xxh3` [15,16], a non-cryptographic hash function, to compute a unique fingerprint for each read. These fingerprints are cached in memory using `fxhash` [17], which provides a low memory footprint.

FDedup provides users with precise control over the hash collision rate while maintaining high performance. It automatically scales its hashing strategy, seamlessly switching between 64-bit and 128-bit hashes based on the estimated input sequence count and a user-defined collision probability threshold.

Specifically designed for seamless integration into OMICS pipelines, the tool features robust incremental deduplication and auto-recovery mechanisms. In the event of an interruption FDedup can safely preload existing hashes from the output file to prevent duplicate processing and smoothly resume operations. If an uncompressed output file becomes corrupted due to a crash, the tool automatically detects the issue, calculates a fail-safe truncation point, and truncates the file to the last valid sequence before continuing.

Finally, a dry-run mode (`--dry-run` or `-s`) is available to calculate the duplication rate without writing any output files. Since file I/O operations account for the majority of execution time, this mode is exceptionally fast and serves as a highly efficient feature for pipeline planning and data analyses.

3.1 : Algorithmic Complexity and Hardware-Level Optimizations

The architectural principal of `FastDedup`'s performance lies in shifting the deduplication problem from a string-matching paradigm to an integer-matching paradigm. By transforming raw paired-end sequences (e.g., 2 x 150 bytes) into a single 64-bit hash, reducing the data by 37.5 times in both space and time complexity.

3.1.1 Space Complexity and Memory Footprint

In a deterministic approach, storing a read pair requires retaining sequences of length L (where $L = r_1 + r_2$). The space complexity for N unique sequences scales linearly as $O(N \times L)$. For a standard 150 bp paired-end dataset, each pair consumes at least 300 bytes of raw character data, plus substantial data structure overhead (e.g., string pointers). Storing 10^8 sequences typically exceeds 32 GB of RAM. By computing a 64-bit hash using `xxh3`, `FastDedup` reduce the sequence identity into exactly 8 bytes. This decouples the memory requirement from the read length, reducing the space complexity to $O(N)$ (since $8 \ll L$).

3.1.2 Time Complexity and Arithmetic Logic Unit (ALU) Efficiency

String-based deduplication necessitates sequential, byte-by-byte comparisons. In the worst-case scenario (ex: highly similar sequences or long identical prefixes), comparing two sequences requires $O(L)$ time. Conversely, while calculating the initial hash takes $O(L)$ time during the single-pass file reading, all subsequent identity verifications are reduced to integer operations. Comparing two 64-bit integers is executed natively by the CPU's ALU in exactly 1 clock cycle, effectively reducing the hash lookup and comparison time complexity to $O(1)$.

3.1.3 Cache Locality and Zero-Allocation Pipeline

Beyond theoretical Big-O improvements, the 8-byte hash architecture heavily exploits modern CPU hardware design. Processors fetch data from RAM in discrete 64-byte cache lines. A 300-byte sequence spans multiple cache lines, leading to frequent cache misses and severe memory bandwidth bottlenecks. In contrast, exactly eight 64-bit hashes fit perfectly into a single cache line. As `fxhash` probes the internal hash set for collisions, it operates almost exclusively within the ultra-fast CPU cache (L1/L2), minimizing latency and maximizing throughput.

Furthermore, `FastDedup` processes byte slices directly from the `needletail` [18] parser into the `xxh3` hasher without ever allocating the raw string to the heap. This zero-allocation pipeline completely bypasses system-level memory allocation (`malloc`) overheads, ensuring that the execution time is strictly bound by disk I/O rather than RAM latency.

3.2 : The Mathematics of Dynamic Hashing

While `FDedup` defaults to the highly efficient 64-bit hashing strategy of `xxh3`, it dynamically assesses the risk of hash collisions to guarantee data integrity. The theoretical risk of a hash collision p for x sequences follows the standard Birthday Problem approximation [19,20]:

$$p = \frac{x^2}{2 * 2^N}$$

Where $N \in \{64, 128\}$ represents the bit-length of the hash. At the default 64-bit configuration and a probability threshold of 0.001, it requires approximately $0.19 * 10^9$ sequences to reach a 1‰ collision risk. If the estimated dataset size pushes the collision probability beyond this threshold, `FDedup` automatically scales to 128-bit hashing, effectively nullifying the risk (requiring $0.28 * 10^{17}$ sequences to reach the same probability).

For paired-end reads, maintaining this theoretical probability requires bitwise operations. Let S_1 and S_2 represent the nucleotide sequences of Read 1 and Read 2, and H be our non-cryptographic uniform hash function mapping to $\{0,1\}^N$. We define the independent hashes for each read as:

$$h_1 = H(S_1)$$

$$h_2 = H(S_2)$$

If the algorithm combined these hashes using a simple bitwise XOR ($\oplus$), a critical failure occurs when Read 1 and Read 2 are identical sequences ($S_1 = S_2$). In this case, $h_1 = h_2$, resulting in:

$$h_c = h_1 \oplus h_1 = 0$$

Every paired-end read with identical forward and reverse sequences would map to 0. This would cause massive, deterministic hash collisions, meaning the collision probability would no longer be bounded by the theoretical uniform distribution, but rather by the biological frequency of identical sequence pairs in the dataset.

To prevent symmetrical cancellation, the algorithm introduces a bitwise left-rotation by $\frac{N}{2}$ bits, denoted as $R(x, k)$. The unified fingerprint h_u is instead calculated as:

$$h_u = h_1 \oplus R\left(h_2, \frac{N}{2}\right)$$

Under this operation, even if $S_1 = S_2$:

$$h_u = h_1 \oplus R\left(h_1, \frac{N}{2}\right) \neq 0$$

Because $R\left(h_1, \frac{N}{2}\right)$ shifts the lower half of the bits to the upper half and vice versa, $h_1 \oplus R\left(h_1, \frac{N}{2}\right)$ will almost never equal 0. It would only equal 0 if h_1 consisted of two perfectly identical halves, an event with an astronomically low probability of $2^{-\frac{N}{2}}$.

Since a bitwise rotation is a bijective function over the N-bit vector space, it perfectly preserves the original entropy of h_2. When two independent, uniformly distributed random variables are combined using a bitwise XOR, the resulting variable remains uniformly distributed. Therefore, the unified fingerprint h_u safely maintains a strict uniform distribution, ensuring the overall collision risk remains uncorrupted and strictly bounded by the initial probability formula.

This is the mathematical foundation that allows `FDedup` to guarantee a collision probability of less than 1‰ for datasets containing up to approximately 190 million paired-end reads (380 million total sequences) when using 64-bit hashes.

3.3 : Architecture

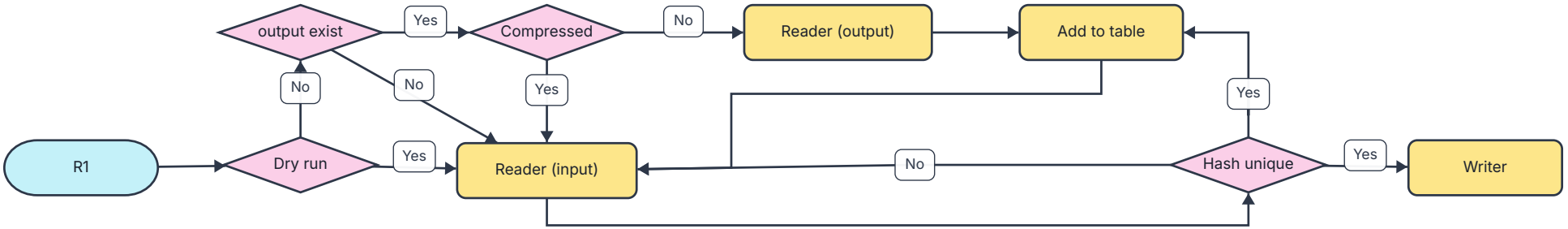


Figure 2: Architectural design of `FastDedup` for Single-End (SE) reads.

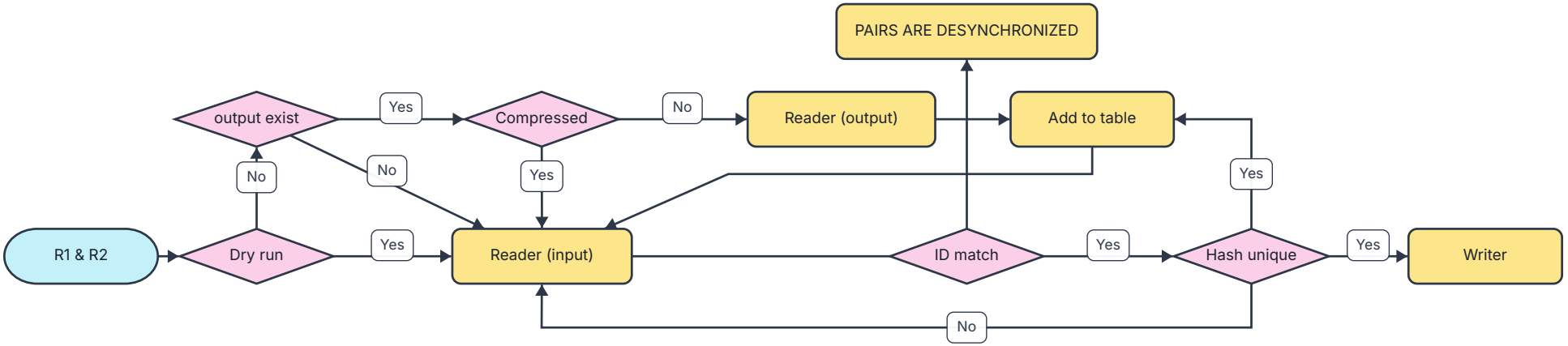


Figure 3: Architectural design of `FastDedup` for Paired-End (PE) reads.

4 – Benchmarking and performance evaluation

To test the performances of the differents tools evoked in the previous sections (Table 1), I benchmarked them on a series of synthetic datasets of varying sizes, ranging from 100,000 to 300 million paired-end reads to wich I added 10% of duplicates to each dataset. This allows us to evaluate the scalability of each tool and its suitability for different types of sequencing projects, from small-scale experiments to large population genomics studies. To run those benchamrks in a controled environment, I set-up a snakemake pipeline with each tool having 1 core and 33GB of RAM available. This pipeline ran on the High-Performance Computing (HPC) cluster I/O of Meso with AMD EPYC 9654 processors. If the tool goes above 32GB of RAM, it is killed and written as OOM (Out Of Memory) in the results table. In times where RAM and electricity are becoming increasingly costly and environmentally impactful, it is crucial to consider the resource efficiency of bioinformatics tools. Since those tools are for the first step for pre-treatment of data, it would be senseless to run them on luducrous ressources.

4.1 : Compressed Paired-End FASTQ

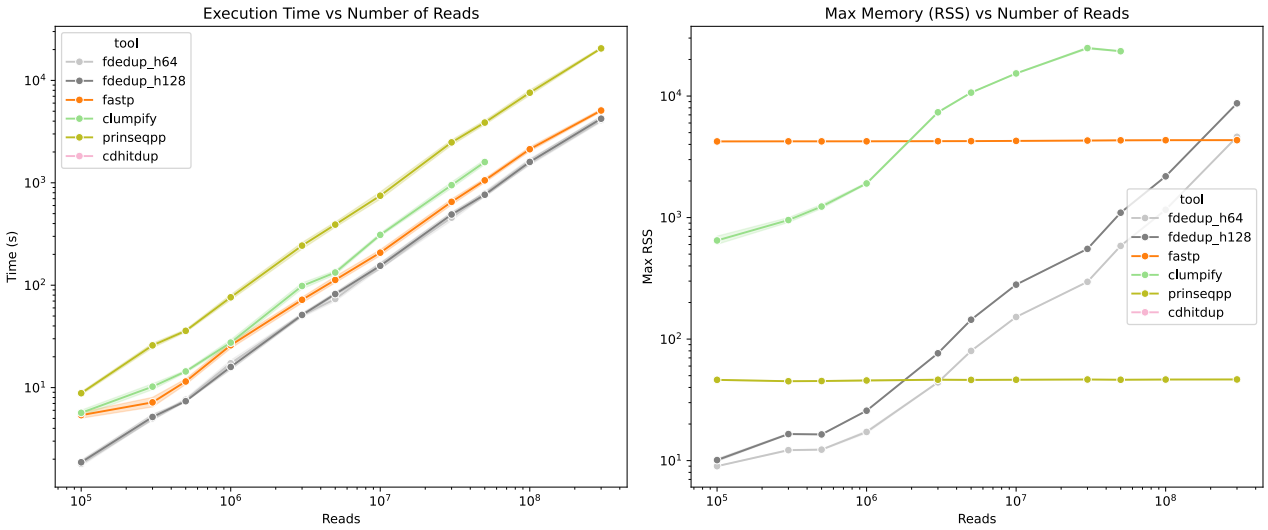


Figure 4: Execution time and memory usage across varying dataset sizes.

The default compression level of `fastp` is 4 but the `flate` library I use with `fdedup` is using a level 6 compression by default, which is more CPU intensive. Adding an option to set the compression level, we can now put `fastp` and `fdedup` on a more equal footing in terms of compression time. It would have been easier to just setp fastp to level 6, but it was a nice opportunity to add another feature to `fdedup`. In the end I added the option but I will rerun the benchmarks with `fastp` set to level 6 because all the other tools are level 6.

4.2 : Uncompressed Paired-End FASTQ

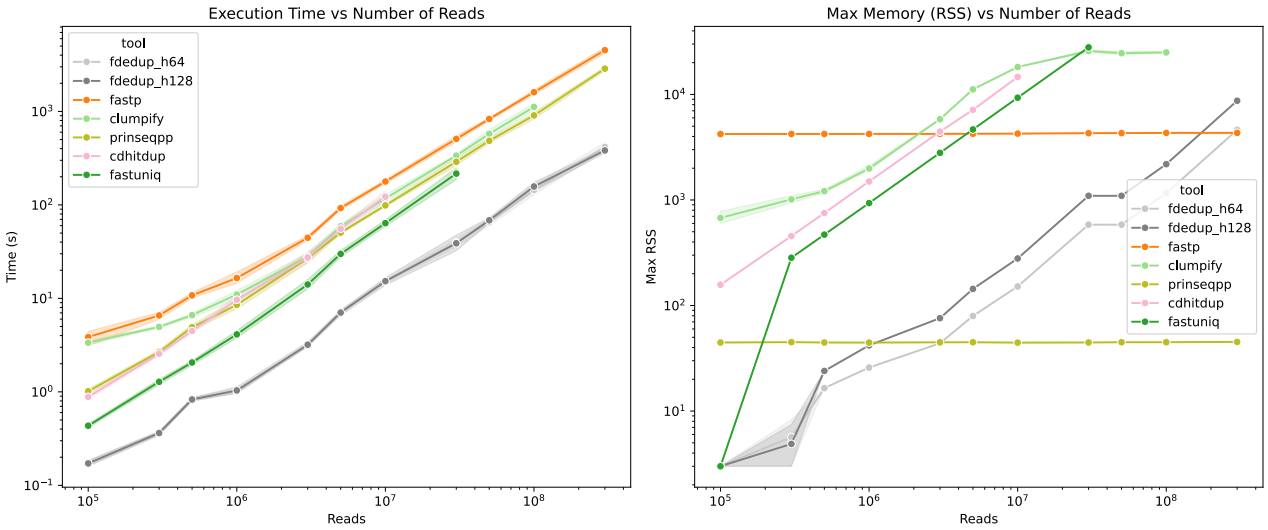


Figure 5: Execution time and memory usage across varying dataset sizes.

For uncompressed FASTQ file as inputs and uncompressed FASTQ file as outputs, `FastDedup` consistently outperformed all other tools in terms of execution time (Figure 5, Table 3). We can also observed that there are no real difference in execution time between the 64-bit and 128-bit hashing strategies, which is consistent with the theoretical analysis of their time complexity. We can also see that `FastDedup` takes up significantly less memory than all the other tools up until 5M reads, where `prinseq++` starts to use less memory than `FastDedup`.

However, starting at 30M reads, `CD-HIT-DUP` crashed due to an OOM error, followed by `fastuniq` at 50M reads, and `clumpify` at 300M reads. This left only `fastp`, `prinseq++` and `FastDedup` which was 6.4 times faster than `prinseq++` and 11.3 times faster than `fastp` at 300M reads, while using only 4.6GB and 8.7GB of RAM for 64-bit and 128-bit hashing respectively.

We can also see that the difference of RAM usage bewteen the 64-bit and 128-bit hashing strategies is doubled, which is consistent with the the fact that 64 times 2 is 128.

4.3 : Compressed in, decompressed out Paired-End FASTQ

Before talking about the results, we have to note that `fastuniq` is not included in this benchmark since it does not support compressed files. For `CD-HIT-DUP`, I got OOM errors for all read volumes. Since the tool doesn't output any log, I wasn't able to make it wrork for this specific benchmark.

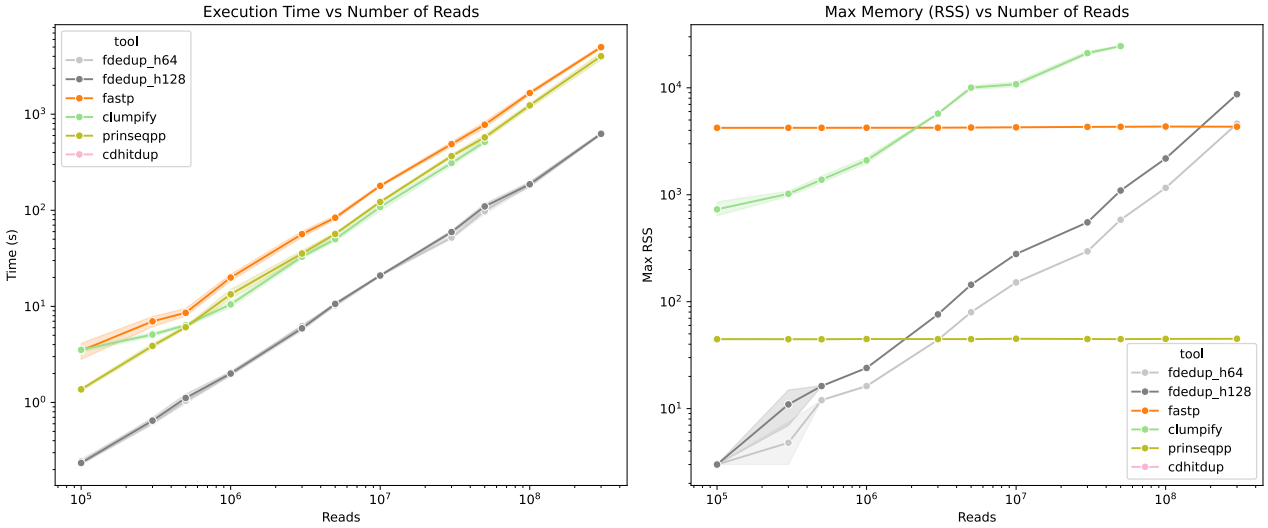


Figure 6: Execution time and memory usage across varying dataset sizes.

We observe very similar results as for uncompressed files (Figure 6, Table 4). `FastDedup` being the fastest tool and using the least amount of RAM up until 5M reads, where `prinseq++` starts to use less memory than `FastDedup`. We can also note very similar results for relative performance between each tools, which mean that all the tools scale in a similar way when the input files are compressed or not.

4.4 : Compressed Single-End FASTQ

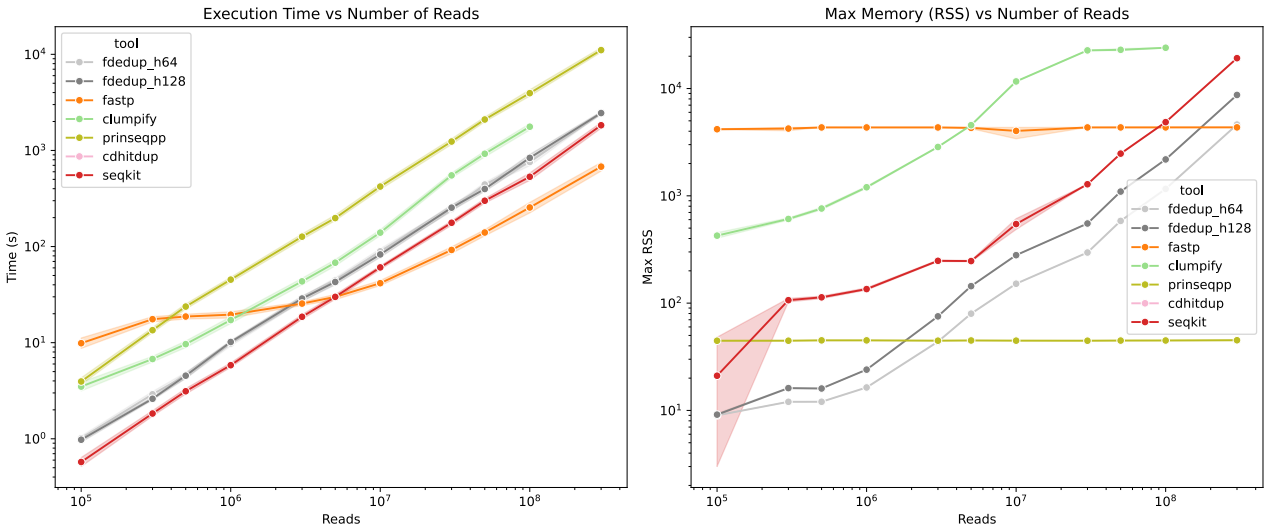


Figure 7: Execution time and memory usage across varying dataset sizes.

For SE data, we see 2 clear winners for execution time, `SeqKit` up until 3M reads and `fastp` for 5M reads and above (Figure 7, Table 5). Despite that `FastDedup` is having much better RAM efficiency than `SeqKit` and `fastp` while keeping a relatively good execution time, especially compared to `prinseq++` which is the slowest tool for SE compressed data.

4.5 : Decompressed Single-End FASTQ

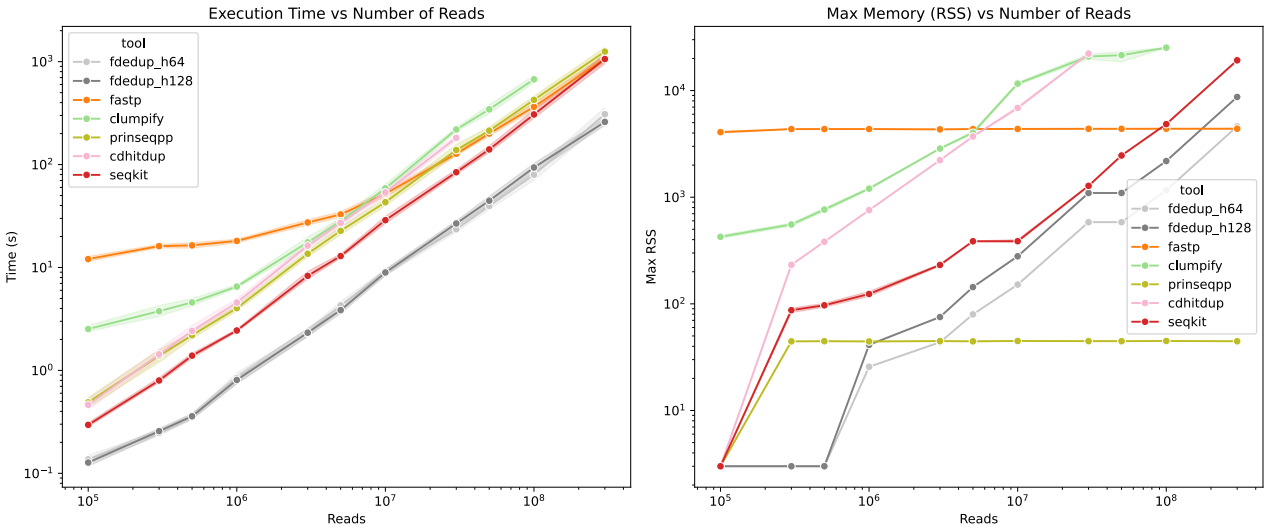


Figure 8: Execution time and memory usage across varying dataset sizes.

Here we get similar results as for uncompressed paired-end data, with `FastDedup` being the fastest tool and using the least amount of RAM up until 5M reads, where `prinseq++` starts to use less memory than `FastDedup` (Figure 8, Table 6).

4.6 : Compressed in, decompressed out Single-End FASTQ

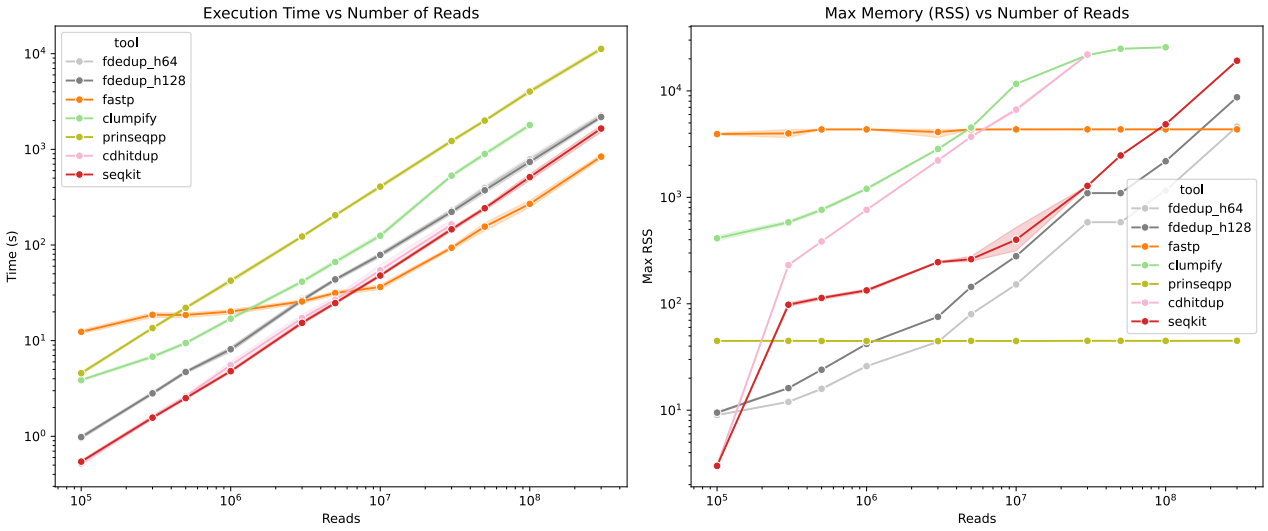


Figure 9: Execution time and memory usage across varying dataset sizes.

Well I fucked up the config of this one it is reruning right now.

5 – Discussion and future perspectives

Now that we have significant data from isolated benchmarks, we can discuss the implications of these results for real-world applications and the future development of `FastDedup`.

5.1 : Output Quality and Deduplication Accuracy

Despite each tool having very different approaches to deduplication, they all produced the pretty much the same output in terms of the number of unique sequences and the duplication rate. There is a small difference, but less than 0.1% between the tools. However, I wasn't able to estimate for `CD-HIT-DUP` since it doesn't output any log. Moreover, there is a problem with `prinseq++` starting at 10M reads where the duplication rate is higher than 10.1%, reaching up to more than 90% at 300M reads (Figure 10). Because of this we do not

recommend using `prinseq++` in your pipelines, since it starting to be more memory efficient than `FastDedup` at 5M reads, but starts having a very high false-positive rate at 10M reads.

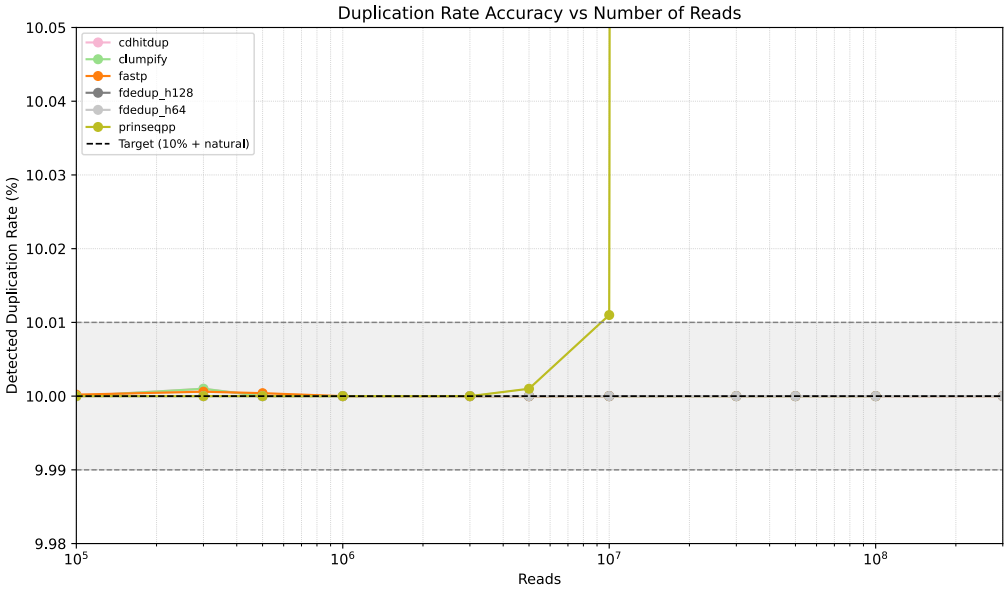


Figure 10: Execution time and memory usage across varying dataset sizes.

A similar pattern is observed for `clumpify` on signle-end data, but starting at 3M reads. It is above 10.1% at 10M reads but reaching around than 10.1% at 100M reads (Figure 11). This is still above the 0.1% false-positive threshold, but it is far better than the 90% false-positive rate of `prinseq++` at 300M reads.

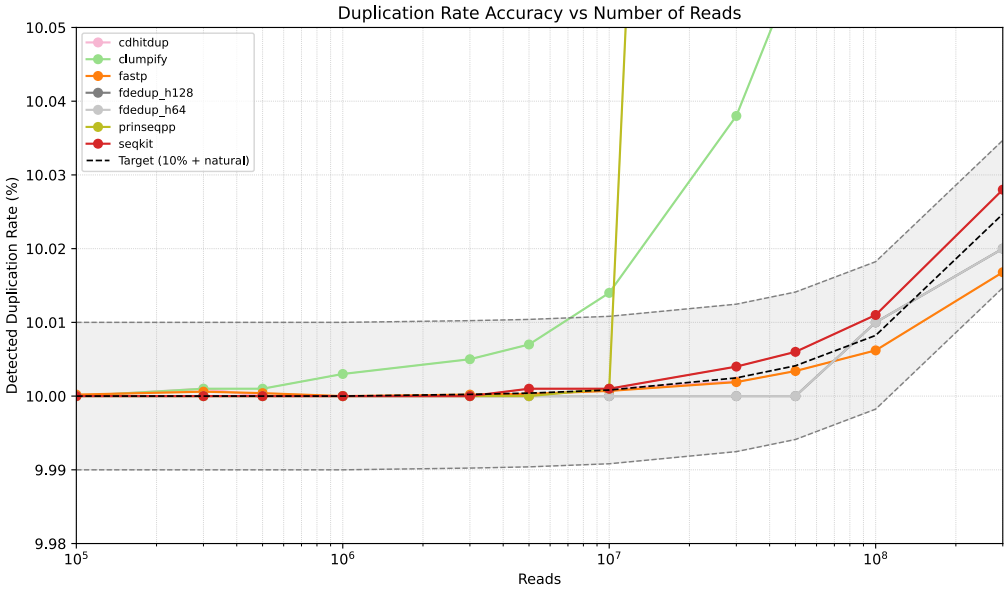


Figure 11: Execution time and memory usage across varying dataset sizes.

The target duplication value isn't constant across different volume because `wgsim` is a random read simulator. We can calculate the number of natural duplicate C with N randomly generate reads from a uniform distribution of M possible start positions like $C \approx \frac{N^2}{2M}$. However, the human genome is not uniform, it contains high-frequency repeat families [21], so we have to aproximate M with the number of unique start positions in the simulated dataset. We can aproximate $k = (2G)^{-1}$ using a known data point from your `wgsim` simulation. Looking at the generation of $270,000,000$ single-end reads, the simulation naturally produced roughly $60,000$ coincidental duplicates.

We can solve k like:

$$60,000 = k \times (270,000,000)^2$$
$$k = \frac{60,000}{(270,000,000)^2}$$

We can simply now express the number of natural duplicates C like $C = k \times N^2$ and add this C to the 10% of duplicates we added to the dataset to get the expected duplication rate for each dataset size.

5.2 : Performance with Paired-End Reads

On all paired-end datasets, `FastDedup` outperformed all other tools in terms of execution time for all dataset sizes, while maintaining a very low memory footprint. However if the user is looking for the lowest possible memory usage, starting after 3M reads, `prinseq++` is the best choice for paired-end data, but it is more than 7 times slower than `FastDedup` for 5M reads and goes down to 6.4 times slower at 300M reads.

5.3 : Performance with Single-End Reads

5.4 : Performance with Compressed Files

5.5 : In short

Tool	Algorithm / Approach	PE Support	Error Tolerance	Memory Usage	Key Limitations
`fastp` [3,4]	Probabilistic (Bloom filter/hashes)	Yes	No	Configurable	False-positive rate; long runtime
`FastUniq` [5]	Deterministic (Merge sort)	Yes	No	Very High	Severe RAM bottleneck; sensitive to input file order.
`Clumpify` [7]	k-mer grouping	Yes	Yes	Variable	Variable stability on massive datasets; high memory overhead.
`SeqKit rmdup` [8,9]	Exact matching (Hashing/MD5)	No	No	Very Low	Desynchronizes paired-end files.
`CD-HIT-DUP` [10,11]	Greedy incremental clustering	Yes	Yes	Very High	Stability issues in containerized environments.
`PRINSEQ++` [12]	Probabilistic (Bloom filter)	Yes	No	Low	False-positives rate
`FastDedup`	Exact matching (Hashing/xxh3)	Yes	No	Very Low	Theoretical negligible collision risk

Table 1: Summary of existing tools for reads deduplication and their limitations.

5.6 : The Future of FastDedup

The current version of `FastDedup` is optimized for speed and memory efficiency, but there are some features that could further enhance its functionality. Firstly, the support for multithreading, which would allow the tool to take advantage of multiple CPU cores and further reduce execution time, especially for larger datasets. We can use thoses mutiple cores for parallelizing the file reading and hashing process, while maintaining a shared hash set for deduplication. This can however introduce some overhead due to thread synchronization, so it would require careful testing to ensure that the performance benefits outweigh the costs. However, those extra cores could be used for faster file compression and decompression. Indeed, writing a compressed file makes the writing phase taking more than 90% of the execution time.

Another feature that could be added is track of abundance, which would allow users to not only remove duplicates but also to quantify how many times each unique sequence was observed in the input dataset. This can be particularly useful for applications like metagenomics or transcriptomics, where the abundance of certain sequences can provide important biological insights.

6 – Conclusion and Recommendations

`FastDedup` represents a significant advancement for FASTX deduplication, offering unparalleled speed and memory efficiency compared to existing tools. Its innovative use of non-cryptographic hashing and dynamic scaling strategies allows it to handle massive datasets with ease, making it an ideal choice for modern high-throughput sequencing projects. While other tools may offer specific features or error tolerance, `FastDedup` provides a robust, scalable solution for bioinformaticians seeking to optimize their data processing pipelines without sacrificing accuracy or resource efficiency. For paired-end datasets, `FastDedup` is the clear choice for users prioritizing speed. For single-end datasets, TODO.

We recommand using `FastDedup` at the beginning of the data processing pipeline, immediately after quality control and adapter trimming steps, to ensure that downstream analyses are not biased by PCR duplicates. By using uncompressed output files, users can further maximize the performance benefits of `FastDedup`, and then deleting the uncompressed files after further processing to save storage space.

This software is protected by the MIT License, and you can find it on:
- GitHub https://github.com/RaphaelRibes/FDedup.git
- Bionconda https://anaconda.org/bioconda/fdedup
- Cargo https://crates.io/crates/fastdedup

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8 – Annexes

Tool *(s)* *(MB)*	**0.1M**	**0.3M**	**0.5M**	**1.0M**	**3.0M**	**5.0M**	**10.0M**	**30.0M**	**50.0M**	**100.0M**	**300.0M**
`cdhitdup`	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM
`clumpify`	5.7±0.6 648±84	10.2±1.0 954±72	14.4±0.6 1232±78	27.5±2.7 1909±48	98.3±10.6 7351±122	133±9.1 10671±26	310±18.2 15352±128	949±102 24814±672	1595±145 23383±133	OOM	OOM
`fastp`	5.4±0.6 4233±1	7.2±1.3 4238±1	11.5±1.1 4238±2	25.9±2.8 4240±2	72.0±7.7 4252±6	113±12.8 4257±9	208±22.6 4275±18	652±68.4 4303±22	1056±76.7 4322±31	2128±163 4333±20	5083±252 4336±17
`fdedup_h128`	1.9±0.1 10±0	5.2±0.5 17±0	7.4±0.2 16±0	15.9±1.6 26±0	51.2±2.9 76±0	82.0±4.6 144±0	155±18.0 281±0	490±43.7 552±0	765±59.7 1097±0	1599±130 2184±0	4226±410 8713±0
`fdedup_h64`	1.8±0.1 9±0	5.0±0.2 12±0	7.6±0.4 12±0	17.3±1.9 17±1	50.7±3.2 44±0	73.4±6.9 80±0	154±14.8 152±1	457±48.9 296±1	782±66.9 585±1	1599±188 1160±1	4101±387 4617±0
`prinseqpp`	8.8±0.5 46±1	25.9±2.1 45±0	35.8±1.8 45±0	76.2±7.0 46±1	244±28.1 46±0	391±36.0 46±1	748±99.2 46±0	2482±222 47±0	3875±294 46±0	7584±720 46±0	20538±720 47±0

Table 2: Execution time and memory usage across varying dataset sizes for compressed input and compressed output paired-end FASTQ files.

Tool *(s)* *(MB)*	**0.1M**	**0.3M**	**0.5M**	**1.0M**	**3.0M**	**5.0M**	**10.0M**	**30.0M**	**50.0M**	**100.0M**	**300.0M**
`cdhitdup`	0.9±0.1 157±1	2.6±0.2 455±3	4.5±0.3 752±0	9.7±1.8 1497±1	27.4±6.0 4443±1	55.6±5.1 7146±118	122±26.8 14639±1	OOM	OOM	OOM	OOM
`clumpify`	3.4±0.3 676±155	5.0±0.2 1013±131	6.6±0.5 1211±85	11.0±2.1 1989±181	28.0±3.8 5825±83	59.1±6.0 11166±35	117±17.6 18191±21	337±19.9 25774±1431	576±105 24599±1394	1118±147 25008±829	OOM
`fastp`	3.9±1.1 4217±1	6.6±0.9 4219±1	10.8±1.1 4222±1	16.5±3.8 4225±3	44.5±4.1 4230±3	92.5±7.5 4239±7	178±14.0 4254±10	508±54.2 4292±15	831±50.3 4302±17	1606±149 4319±24	4522±527 4323±18
`fastuniq`	0.4±0.0 3±0	1.3±0.1 284±0	2.1±0.2 470±0	4.1±0.6 935±0	14.1±2.7 2796±0	29.9±3.8 4658±0	63.8±8.0 9312±0	216±51.7 27927±0	OOM	OOM	OOM
`fdedup_h128`	0.2±0.0 3±0	0.4±0.0 5±4	0.8±0.1 24±0	1.0±0.1 42±0	3.2±0.3 76±0	7.0±0.8 144±0	15.3±2.2 279±0	38.8±13.8 1096±1	68.5±6.8 1095±0	157±26.9 2184±1	383±32.8 8712±0
`fdedup_h64`	0.2±0.0 3±0	0.4±0.0 6±4	0.8±0.1 17±0	1.0±0.1 26±0	3.1±0.1 44±0	7.2±0.4 80±0	15.1±1.1 152±0	39.7±6.6 584±0	65.5±8.9 584±0	145±29.5 1160±0	418±65.7 4616±0
`prinseqpp`	1.0±0.1 45±0	2.7±0.3 45±0	4.9±0.5 45±1	8.5±1.6 45±0	26.8±5.3 45±0	50.5±2.9 45±0	98.8±5.6 45±0	290±43.9 45±0	485±45.8 45±0	907±117 45±0	2869±232 45±1

Table 3: Execution time and memory usage across varying dataset sizes for uncompressed input and uncompressed output paired-end FASTQ files.

Tool *(s)* *(MB)*	**0.1M**	**0.3M**	**0.5M**	**1.0M**	**3.0M**	**5.0M**	**10.0M**	**30.0M**	**50.0M**	**100.0M**	**300.0M**
`cdhitdup`	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM
`clumpify`	3.5±0.3 730±189	5.1±0.4 1020±89	6.4±0.4 1383±162	10.5±1.1 2098±240	32.8±3.0 5735±134	50.0±4.2 10040±669	108±13.6 10794±950	310±31.5 21110±1336	515±46.4 24537±428	OOM	OOM
`fastp`	3.5±1.1 4226±1	7.0±1.4 4230±1	8.6±1.3 4230±1	20.0±2.6 4233±1	56.5±6.7 4240±3	83.7±6.4 4251±6	180±9.1 4270±15	489±42.6 4312±25	777±94.3 4324±23	1662±121 4344±13	4986±304 4332±12
`fdedup_h128`	0.2±0.0 3±0	0.6±0.1 11±6	1.1±0.2 16±0	2.0±0.2 24±0	5.9±0.5 76±0	10.6±0.6 144±0	20.9±0.8 279±1	59.4±4.1 552±0	110±12.9 1095±0	186±17.1 2184±0	625±26.8 8712±0
`fdedup_h64`	0.2±0.0 3±0	0.6±0.1 5±4	1.0±0.1 12±0	2.0±0.1 16±0	6.2±0.4 44±0	10.2±0.6 80±0	21.0±0.8 151±0	52.1±3.8 296±0	97.9±10.5 584±0	188±15.7 1160±0	629±24.8 4616±0
`prinseqpp`	1.4±0.1 45±1	3.9±0.4 45±0	6.1±0.3 45±0	13.4±2.4 45±0	35.5±4.1 45±0	56.7±3.3 45±0	122±5.6 45±0	366±25.5 45±0	573±59.4 45±0	1235±112 45±0	4015±592 45±0

Table 4: Execution time and memory usage across varying dataset sizes for compressed input and uncompressed output paired-end FASTQ files.

Tool *(s)* *(MB)*	**0.1M**	**0.3M**	**0.5M**	**1.0M**	**3.0M**	**5.0M**	**10.0M**	**30.0M**	**50.0M**	**100.0M**	**300.0M**
`cdhitdup`	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM	OOM
`clumpify`	3.5±0.6 426±40	6.8±0.8 609±24	9.7±1.2 762±34	17.2±2.2 1202±42	43.4±5.4 2853±21	68.0±8.4 4556±43	140±16.4 11646±53	550±52.8 22673±169	924±103 22955±803	1765±216 23990±87	OOM
`fastp`	9.9±2.1 4177±67	17.6±1.8 4235±257	18.7±1.8 4339±1	19.6±2.3 4338±2	25.5±1.8 4338±1	29.9±3.1 4300±110	41.6±4.2 4027±938	92.2±12.3 4339±1	140±16.6 4341±3	255±52.2 4342±8	678±115 4346±10
`fdedup_h128`	1.0±0.1 9±0	2.6±0.1 16±0	4.5±0.5 16±0	10.2±0.6 24±0	28.8±2.9 75±0	42.8±4.8 144±0	82.8±10.1 280±0	254±28.8 552±0	397±40.2 1096±1	837±123 2184±0	2450±143 8712±0
`fdedup_h64`	1.0±0.1 9±0	2.9±0.3 12±0	4.5±0.5 12±0	9.8±0.9 16±0	28.4±2.4 44±0	44.3±5.0 80±0	89.2±10.1 152±0	246±25.0 296±0	442±40.2 584±0	759±121 1160±0	2457±97.3 4616±0
`prinseqpp`	3.9±0.5 45±0	13.5±1.4 45±0	23.7±2.4 45±0	45.3±4.3 45±0	127±14.3 45±0	198±20.4 45±0	421±49.9 45±0	1238±140 45±0	2099±194 45±0	3944±483 45±0	11065±1007 45±1
`seqkit`	0.6±0.1 21±36	1.8±0.2 107±6	3.1±0.4 113±5	5.8±0.5 135±5	18.6±1.7 248±6	30.0±2.9 247±5	60.7±4.2 546±107	177±15.3 1282±4	300±26.3 2474±6	532±73.3 4862±7	1828±201 19189±16

Table 5: Execution time and memory usage across varying dataset sizes for compressed input and compressed output single-end FASTQ files.

Tool *(s)* *(MB)*	**0.1M**	**0.3M**	**0.5M**	**1.0M**	**3.0M**	**5.0M**	**10.0M**	**30.0M**	**50.0M**	**100.0M**	**300.0M**
`cdhitdup`	0.5±0.1 3±0	1.4±0.2 232±0	2.4±0.5 383±0	4.6±0.6 756±1	16.3±1.4 2224±1	27.1±5.3 3713±12	53.6±7.7 6875±367	182±17.2 22202±220	OOM	OOM	OOM
`clumpify`	2.5±0.3 425±25	3.8±0.7 555±33	4.6±0.6 766±51	6.5±0.4 1204±46	17.6±3.2 2861±21	28.1±3.3 4024±226	58.8±9.2 11618±520	219±15.9 20868±1776	344±52.6 21473±4147	673±92.0 25250±290	OOM
`fastp`	12.1±1.1 4075±149	16.1±1.0 4352±1	16.4±1.7 4356±2	18.1±1.1 4357±4	27.4±2.6 4328±69	32.8±3.4 4359±6	52.0±5.9 4368±12	127±9.3 4380±9	200±19.8 4381±9	364±45.2 4385±6	1074±146 4389±2
`fdedup_h128`	0.1±0.0 3±0	0.3±0.0 3±0	0.4±0.0 3±0	0.8±0.1 41±1	2.3±0.4 75±0	3.9±0.5 144±1	8.9±1.0 279±0	26.8±3.0 1095±0	44.6±7.0 1095±0	93.7±15.2 2183±1	260±34.3 8711±0
`fdedup_h64`	0.1±0.0 3±0	0.2±0.0 3±0	0.4±0.0 3±0	0.8±0.2 26±0	2.3±0.3 44±0	4.3±0.6 80±0	9.0±0.9 152±0	23.5±3.0 583±0	39.6±4.5 583±0	79.7±15.5 1160±0	308±76.0 4615±0
`prinseqpp`	0.5±0.1 3±0	1.4±0.3 44±0	2.2±0.3 45±0	4.0±0.4 44±0	13.6±1.7 45±0	22.7±3.0 45±1	43.1±5.3 45±0	139±27.3 45±1	214±30.6 45±0	426±54.3 45±0	1252±205 45±0
`seqkit`	0.3±0.0 3±0	0.8±0.1 87±7	1.4±0.1 97±7	2.4±0.1 124±11	8.3±1.2 232±8	12.9±1.0 386±3	28.8±4.0 387±20	84.3±7.7 1278±7	140±16.5 2461±6	307±48.0 4847±14	1060±183 19215±23

Table 6: Execution time and memory usage across varying dataset sizes for uncompressed input and uncompressed output single-end FASTQ files.

Tool *(s)* *(MB)*	**0.1M**	**0.3M**	**0.5M**	**1.0M**	**3.0M**	**5.0M**	**10.0M**	**30.0M**	**50.0M**	**100.0M**	**300.0M**
`cdhitdup`	0.5±0.0 3±0	1.6±0.1 231±0	2.6±0.1 385±0	5.5±0.7 763±1	17.2±2.4 2220±1	26.7±3.1 3708±1	54.5±10.8 6671±413	164±27.9 21937±154	OOM	OOM	OOM
`clumpify`	3.9±0.2 413±40	6.8±0.5 583±36	9.4±0.5 763±43	17.0±1.0 1203±48	41.4±2.7 2845±42	66.5±5.5 4522±34	125±10.7 11645±52	530±30.1 21697±177	894±74.6 24870±248	1796±141 25688±305	OOM
`fastp`	12.4±0.9 3930±130	18.6±1.7 3989±561	18.6±1.7 4344±5	20.1±1.8 4350±4	25.7±1.9 4107±542	31.5±2.3 4342±9	36.4±3.5 4349±2	93.6±6.4 4352±4	156±25.0 4350±2	269±37.9 4352±3	839±73.4 4353±1
`fdedup_h128`	1.0±0.1 9±0	2.8±0.2 16±0	4.7±0.2 24±0	8.1±0.7 42±0	26.4±1.9 75±0	43.7±2.3 144±0	78.8±6.2 280±0	222±20.8 1096±1	374±43.4 1096±1	739±88.7 2184±0	2178±169 8712±0
`fdedup_h64`	1.0±0.0 9±0	2.8±0.1 12±0	4.7±0.3 16±0	8.0±0.7 26±0	27.3±1.4 44±0	42.5±1.5 80±0	77.9±6.0 152±0	229±17.4 584±0	395±39.7 584±0	786±89.4 1160±0	2254±220 4616±0
`prinseqpp`	4.6±0.3 45±0	13.5±0.7 45±0	22.0±1.3 45±0	42.4±3.2 45±0	123±7.0 45±0	205±10.2 45±0	407±29.7 45±0	1222±85.0 45±0	1998±101 45±0	4021±326 45±0	11219±805 45±0
`seqkit`	0.5±0.0 3±0	1.6±0.1 98±6	2.5±0.1 113±6	4.8±0.3 134±7	15.4±1.0 246±5	24.7±1.2 262±24	48.0±2.5 400±185	146±10.0 1282±6	242±18.8 2473±9	513±65.1 4856±8	1649±220 19192±35

Table 7: Execution time and memory usage across varying dataset sizes for compressed input and uncompressed output single-end FASTQ files.